Supplementary Information

Thermodynamic-limit inference from quantum measurements with discrete diffusion

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S1 Physical models, measurement datasets and numerical references

Hamiltonians and operator conventions

We write X_i, Y_i and Z_i for Pauli operators, with eigenvalues ± 1, and $\mathbf{S}_i = \boldsymbol{\sigma}_i/2$ for spin-1/2 operators. The one-dimensional spin models use periodic boundary conditions, so that site labels are interpreted modulo N. The transverse-field Ising (TFI) Hamiltonian is

$$H_{\mathrm{TFI}}(g) = -(1-g)\sum_{i=1}^{N} Z_i Z_{i+1} - g\sum_{i=1}^{N} X_i, \qquad 0 \le g \le 1. \tag{S1}$$

Thus, g controls both the field and the exchange coefficient; it is not the field-to-exchange ratio of a Hamiltonian with a fixed exchange coefficient. The frustrated J_1–J_2 chain is defined in the Pauli convention as

$$H_{J_1J_2}(\alpha) = J_1\sum_{i=1}^{N} \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+1} + J_2\sum_{i=1}^{N} \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+2}, \qquad J_1 = 1, \quad \alpha = J_2/J_1. \tag{S2}$$

In a spin-operator implementation, the corresponding exchange coefficients are $4J_1$ and $4J_2$, and a reported Pauli spin-dot correlation is four times the corresponding $\langle \mathbf{S}_i \cdot \mathbf{S}_{i+r}\rangle$. For the axial next-nearest-neighbour Ising (ANNNI) chain, we use

$$H_{\mathrm{ANNNI}}(g,\kappa) = -J_1\sum_{i} Z_i Z_{i+1} + \kappa J_1\sum_{i} Z_i Z_{i+2} - g\sum_{i} X_i, \qquad J_1 = 1, \quad \kappa = 0.4. \tag{S3}$$

Here g denotes the transverse-field coefficient and, unlike Eq. S1, does not follow the interpolating convention of the TFI model.

The two-dimensional TFI model is defined on a rectangular $L_x \times L_y$ periodic lattice, with $N = L_xL_y$,

$$H_{\mathrm{2D}}(g) = -(1-g)\sum_{\langle \mathbf{r},\mathbf{r}'\rangle} Z_{\mathbf{r}} Z_{\mathbf{r}'} - g\sum_{\mathbf{r}} X_{\mathbf{r}}, \tag{S4}$$

where each undirected nearest-neighbour bond is counted once. For the Rydberg chain, $n_i = |r_i\rangle\langle r_i| = (Z_i+1)/2$ denotes the excitation occupation. The physical Hamiltonian convention is

$$\frac{H_{\mathrm{Ryd}}}{\Omega} = \frac{1}{2}\sum_{i=1}^{N} X_i - d\sum_{i=1}^{N} n_i + \sum_{r=1}^{R_{\mathrm{cut}}} V_r \sum_{i=1}^{N-r} n_i n_{i+r}, \qquad d = \frac{\delta}{\Omega}, \quad V_r = \frac{(R_b/a)^6}{r^6}. \tag{S5}$$

The chain has open ends and $R_b/a = 2.35$.

Measurement records and dataset partitions

For the one- and two-dimensional spin models, each record consists of $\mathbf{P} \in \{X,Y,Z\}^N$ and $\mathbf{b} \in \{0,1\}^N$. The bases are sampled independently and uniformly at each site. With $s_i = 2b_i - 1$, the reference probability of a complete outcome record is

$$p(\mathbf{b} \mid \mathbf{P}, \boldsymbol{\lambda}) = \mathrm{Tr}\left[\rho_{N,\boldsymbol{\lambda}} \bigotimes_{i=1}^{N} \frac{I + s_i P_i}{2}\right], \tag{S6}$$

where $\rho_{N,\boldsymbol{\lambda}}$ is the numerical ground-state reference. Sampling uses conditional Born probabilities and state updates, thereby retaining the joint outcome statistics. These records define local-Pauli classical shadows [1]. Rydberg data instead consist of fixed-Z occupation records $\mathbf{n} \in \{0,1\}^N$

sampled from finite matrix product states (MPS); no random Pauli bases are sampled for this dataset. The occupation-to-token conversion is specified in Section S4.

The initial one-dimensional training sizes are $N \in \{8, 10, 12, 14\}$. The training grids are

$$\mathcal{G}_{\mathrm{TFI}} = \mathcal{G}_{J_1 J_2} = \{0, 0.25, 0.4, 0.5, 0.6, 0.75, 0.9, 1.0\}, \tag{S7}$$

$$\mathcal{G}_{\mathrm{ANNNI}} = \{0, 0.25, 0.4, 0.5, 0.6, 0.75, 0.9, 1.0, 1.25, 1.5, 1.75, 2.0\}. \tag{S8}$$

These grids refer to g, α and g, respectively. Subsequent recursive training uses the retained exact and previously generated datasets described in main-text Methods. The two-dimensional training set contains the six geometries 3×3, 3×4, 3×5, 4×3, 4×4 and 5×3. The same trained two-dimensional model is directly evaluated on 6×6 and 7×7 lattices. Rydberg training uses $N \in \{16, 24, 32\}$, with detunings extending up to $d = 5.0$.

The TFI evaluations on thermodynamic-limit inference use 15 uniformly spaced values of g in $[0, 1]$. The one-dimensional evaluation grids for the J_1–J_2 and ANNNI models also contain 15 uniformly spaced values in $[0, 1]$ and $[0, 2]$, respectively. The evaluation of two-dimensional TFI model is conducted on 8 g values in {0,0.25,0.4,0.5,0.6,0.75,0.9,1.0}. The standard one-dimensional generated ensembles contain $M = 10{,}000$ records per size and parameter point; the alternative budgets in Section S7 are exceptions. This generated-record count is distinct from the number of records used for training.

Numerical reference calculations and data generation

Finite spin systems are treated by exact diagonalization (ED). For the two-dimensional TFI lattices, the Hamiltonian is represented in matrix-free form and its ground state is obtained using a sparse Lanczos eigensolver with tolerance 10^{-10} and a maximum of 10,000 iterations [2]. The calculation is performed in the full computational basis without explicit symmetry-sector reduction, using a normalized uniform positive vector as the initial state. The resulting ground states are used both to generate projective-measurement records and to evaluate finite-size observables.

For one-dimensional TFI evaluation, systems with $N \leq 10$ are solved by dense Hermitian diagonalization, whereas larger systems use sparse diagonalization. The resulting density matrix is symmetrized under the global spin flip,

$$\rho_{\mathrm{sym}} = \frac{\rho + P\rho P^\dagger}{\mathrm{Tr}(\rho + P\rho P^\dagger)}, \qquad P = X^{\otimes N}. \tag{S9}$$

This averaging is performed after diagonalization and does not project the state onto a particular parity sector. No subsequent spin-flip averaging is applied in this size range. At the exactly degenerate $g = 0$ endpoint, the finite-size ZZ observables considered here are independent of the choice of state within the ferromagnetic ground-state manifold.

For finite-size J_1–J_2 and ANNNI calculations, systems with $N < 12$ are solved by full diagonalization, whereas larger systems are treated using sparse `eigsh` with a tolerance of 10^{-10}. Parameter sweeps use the converged ground state at the preceding parameter value as the initial vector for the next calculation. At the J_1–J_2 endpoint $J_2 = 1$, where the lowest levels become numerically degenerate, twelve low-energy eigenstates are computed. States within $10^{-8} \max(1, |E_0|)$ of the lowest energy are treated as the degenerate subspace, and the preceding ground-state vector is projected onto this subspace and renormalized to define the reference state. Measurement-data generation uses the same sparse solver; for the J_1–J_2 chain, the ground-state branch is followed using intermediate parameter steps of 0.01. All one-dimensional ED calculations use the full computational basis without explicit symmetry-sector reduction.

Thermodynamic-limit references for the frustrated and competing-interaction chains are obtained using infinite DMRG (iDMRG) [3, 4]. The calculations use a two-site unit cell and two-site updates with a mixer, without conserved spin quantum numbers. The singular-value truncation threshold is 10^{-12}, the energy convergence threshold is 10^{-10}, and at most 200 sweeps are performed. The nearest-neighbour J_1–J_2 reference averages the two bond parities of the unit cell. ANNNI reference normalization and the treatment of the disconnected contribution are described in Section S3.

For the TFI chain, thermodynamic-limit ZZ correlations are obtained from the exact free-fermion solution [5], using the Hamiltonian convention

$$H = -(1-g)\sum_i Z_i Z_{i+1} - g\sum_i X_i. \tag{S10}$$

The corresponding fermionic contractions are evaluated by adaptive quadrature over momentum, and correlations at separation r are computed from the associated $r \times r$ Toeplitz determinant. The endpoints are evaluated analytically,

$$C^{\infty}_{ZZ}(r,0) = 1, \qquad C^{\infty}_{ZZ}(r,1) = 0 \quad (r > 0). \tag{S11}$$

These values provide the thermodynamic-limit references used in Fig. 2.

For the 6×6 and 7×7 two-dimensional TFI tests, finite-lattice ZZ correlations are evaluated using stochastic series expansion (SSE) quantum Monte Carlo (QMC) [6]. The simulations use the same periodic torus geometry and Hamiltonian convention as the ED calculations. Correlations are evaluated for displacements $(0,1)$, $(1,0)$ and $(0,2)$, corresponding to horizontal nearest neighbours, vertical nearest neighbours and horizontal separation two, and are averaged over equivalent lattice pairs. Each parameter point is simulated at inverse temperature $\beta = 128$, using 20,000 thermalization sweeps followed by 100,000 measurement sweeps grouped into 40 bins. The reported SSE values are finite-β thermal expectation values and are used as numerical references for the large-lattice tests.

The Rydberg MPS calculations provide projective-measurement records in the occupation basis and the conditional probabilities entering the Bayes reference. Their numerical settings and contraction checks are described in Section S4.

S2 Implementation and training configurations

The measurement records described in Section S1 are used to train separate Diffushadow models for the different Hamiltonians. Here we describe the token representations, positional encodings, masking procedure and training configurations underlying the architecture in main-text Methods, Sections 4.2 and 4.3.

Token representations and positional encoding

For the one-dimensional models, each measurement record is represented as the interleaved sequence

$$[\lambda, P_1, b_1, \ldots, P_N, b_N],$$

where λ denotes the Hamiltonian control parameter, P_i the measurement basis and b_i the corresponding outcome. The control parameter is mapped to a learned embedding through an affine projection. Separate embeddings are used for the measurement bases and for the outcome categories 0, 1 and [MASK], together with token-type embeddings that distinguish parameter, basis and outcome tokens. Rotary positional embeddings (RoPE) are applied according to the

interleaved sequence positions, with the parameter token at position zero and the basis and outcome tokens of site i at positions $2i-1$ and $2i$, respectively.

The two-dimensional implementation retains the basis and outcome representations but assigns each site its explicit lattice coordinates. Tokens are arranged in row-major order, while their spatial positions are specified explicitly by the two-dimensional coordinates. The control parameter is projected to a vector c_λ and added to the basis and outcome embeddings,

$$h^P_{r,c} = e_P(P_{r,c}) + e_{\text{type}}(P) + c_\lambda, \qquad h^b_{r,c} = e_b(\widetilde{b}_{r,c}) + e_{\text{type}}(b) + c_\lambda,$$

where (r, c) denotes the lattice coordinate and $\widetilde{b}$ is the partially masked outcome configuration. The basis and outcome tokens associated with the same site share the same row and column coordinates.

Spatial information is introduced through axial two-dimensional RoPE. For each attention head, the query and key representations are divided between the two lattice axes. The two components are rotated according to the row and column coordinates and concatenated, for example

$$\widetilde{q}_{r,c} = [R_r q^{\text{row}}_{r,c}; R_c q^{\text{col}}_{r,c}],$$

with the same construction applied to the keys. This provides explicit two-dimensional positional information independently of the row-major storage order. A linear output layer applied to the outcome-token representations predicts the two possible measurement outcomes at each site.

Model architecture

Diffushadow uses full self-attention Transformer blocks with rotary positional embeddings, residual connections, layer normalization and GELU feed-forward networks of width $4d$. The one-dimensional TFI, J_1–J_2 and ANNNI models use hidden dimension $d = 256$, six Transformer blocks and eight attention heads. Model-specific architectural and optimization settings for the reported experiments are summarized in Table 1.

Masking and mixed-size training

Training follows the masked-outcome objective described in main-text Methods, Section 4.3. For each record, a masking probability is sampled and outcome sites are masked independently according to that probability. Masks are resampled during training, while the Hamiltonian control parameter and measurement bases remain visible. Cross-entropy loss is evaluated only on masked outcomes and averaged over the masked sites in each mini-batch. The masking ranges used for the reported models are listed in Table 1.

Mixed-size datasets are trained without padding records of different system sizes into a common sequence length. One-dimensional records are grouped by system size and two-dimensional records by lattice geometry, so that each mini-batch contains a common token length and, for the two-dimensional case, a common lattice shape. Mini-batches from the different groups are shuffled during each epoch. Batch sizes are adjusted across system sizes to keep the approximate number of input tokens per optimization step comparable.

Optimization and model selection

Models are optimized using AdamW. The learning rate, batch size, number of epochs, learning-rate schedule and positional-scaling settings used for each experiment are reported in Table 1. When validation-based model selection is used, the checkpoint with the lowest masked-outcome

validation loss is retained for evaluation. The recursive training procedure and the relation between models and generated datasets across successive rounds are described in main-text Methods, Section 4.5.

Supplementary Table 1 | Training configurations for the reported Diffushadow models. d, L and H denote the hidden dimension, number of Transformer blocks and number of attention heads, respectively. η is the initial learning rate, B_{base} is the batch size at the largest training size and E is the number of training epochs.

System	d	L	H	η	B_{base}	E	ϵ	LR schedule	RoPE scaling
1D TFI	256	6	8	1e-5	128	120	0.1	cosine warmup	4.0
J_1–J_2	256	6	8	1e-5	128	120	0.1	cosine warmup	4.0
ANNNI	256	6	8	1e-5	128	120	0.1	cosine warmup	4.0
2D TFI	256	6	8	5e-5	128	120	0.1	cosine warmup	4.0
Rydberg	128	4	8	1e-5	128	100	0.1	cosine warmup	4.0

S3 Classical-shadow observable estimators

Inverse-channel estimators and aggregation

For a uniformly sampled local-Pauli record $(\mathbf{P}_t, \mathbf{b}_t)$, let $s_{t,i} = 2b_{t,i} - 1$. The inverse-channel snapshot and its single-site Pauli estimator are [1]

$$\widehat{\rho}_t = \bigotimes_{i=1}^{N} \left(3\frac{I + s_{t,i}P_{t,i}}{2} - I \right), \qquad \widehat{\sigma}^{\alpha}_{t,i} = 3s_{t,i}\mathbf{1}[P_{t,i} = \alpha]. \tag{S12}$$

For a Pauli string $W = \bigotimes_i W_i$ with support $\mathcal{I} = \{i : W_i \neq I\}$, its single-record estimator is

$$\widehat{W}_t = \mathrm{Tr}(W\widehat{\rho}_t) = \prod_{i\in\mathcal{I}} 3s_{t,i}\mathbf{1}[P_{t,i} = W_i]. \tag{S13}$$

Identity factors equal one. Products involving the same site are first reduced using the Pauli algebra; in particular, $Z_i^2 = I$ is estimated as one.

Spatial averaging is performed within each record. For a translated observable, denote the resulting per-record value by u_t. The records are divided into $K = 10$ equal consecutive groups $\mathcal{B}_k$, and median-of-means aggregation gives

$$\overline{u}_k = \frac{1}{|\mathcal{B}_k|} \sum_{t\in\mathcal{B}_k} u_t, \qquad \widehat{u}_{\text{MoM}} = \mathrm{median}_{k=1,\ldots,K}\, \overline{u}_k. \tag{S14}$$

All record budgets reported here are divisible by K. For even K, we use the lower of the two central group means; for $K = 10$, this corresponds to the fifth ordered group mean. The same estimators are applied to exact and generated measurement ensembles. Direct ED or MPS expectation values are evaluated from the reference states. Snapshot bootstrap resampling is performed on whole records, retaining each record's basis and outcome sequences together.

Spin correlations and two-dimensional displacements

For a periodic chain and nonzero separation r, define the group estimate

$$\widehat{C}^{(k)}_{\alpha\alpha,N}(r) = \frac{1}{|\mathcal{B}_k|N} \sum_{t\in\mathcal{B}_k} \sum_{i=1}^{N} \widehat{\sigma}^{\alpha}_{t,i}\widehat{\sigma}^{\alpha}_{t,i+r}. \tag{S15}$$

The reported TFI ZZ correlation is the MoM of these group estimates for $\alpha = Z$. It is the unconnected correlation $N^{-1}\sum_i \langle Z_i Z_{i+r}\rangle$, with no subtraction of single-site magnetizations. The J_1–J_2 spin-dot estimate is

$$\widehat{C}_{\mathrm{spin},N}(r) = \sum_{\alpha=X,Y,Z} \mathrm{median}_k\, \widehat{C}^{(k)}_{\alpha\alpha,N}(r). \tag{S16}$$

Thus, each spin component is therefore aggregated before summation over components, rather than applying the median after the component sum.

On the two-dimensional torus, the analogous per-record estimator is

$$u_t(\boldsymbol{\Delta}) = \frac{1}{L_x L_y}\sum_{x=0}^{L_x-1}\sum_{y=0}^{L_y-1} \widehat{Z}_{t,(x,y)}\widehat{Z}_{t,(x+\Delta_x,y+\Delta_y)}, \tag{S17}$$

where both coordinates are periodic. The three reported displacements are $(0,1)$, $(1,0)$ and $(0,2)$: nearest neighbours in the two lattice directions and separation two along the second coordinate. The generated records and SSE reference use these same spatial averages.

ANNNI structure factors

The finite-size longitudinal structure factor is

$$S_{Z,N}(q) = \frac{1}{N}\sum_{i,j=1}^{N} \cos[q(i-j)]\langle Z_i Z_j\rangle = \sum_{r=0}^{N-1}\cos(qr)C_{ZZ,N}(r), \tag{S18}$$

for the commensurate momenta used here, $q = \pi$ and $q = \pi/2$. The reported $q = \pi/2$ comparisons use sizes divisible by four. There is no further division by N. The self-correlation is $C_{ZZ,N}(0) = 1$. In the measurement estimator, the Fourier sum is formed *inside each group* before MoM aggregation:

$$\widehat{S}^{(k)}_{Z,N}(q) = 1 + \sum_{r=1}^{N-1}\cos(qr)\widehat{C}^{(k)}_{ZZ,N}(r), \qquad \widehat{S}_{Z,N}(q) = \mathrm{median}_k\, \widehat{S}^{(k)}_{Z,N}(q). \tag{S19}$$

This procedure is not a Fourier sum of separately aggregated correlations. The finite-size observable contains the full, unconnected correlation. For a translation-invariant state and the nonzero commensurate momenta above, the uniform disconnected contribution vanishes exactly.

The infinite-MPS reference is evaluated from connected correlations $C^c(r) = \langle Z_0 Z_r\rangle - \langle Z_0\rangle\langle Z_r\rangle$. The infinite-MPS reference uses a one-sided correlation sum truncated at $r = 199$. For comparison with the two-sided convention used in the figures, we evaluate

$$S^{\mathrm{ref}}_Z(q) = C^c(0) + 2\sum_{r=1}^{199}\cos(qr)C^c(r) = 2S^{\mathrm{stored}}_Z(q) - C^c(0). \tag{S20}$$

where $C^c(0) = 1 - m_z^2$.

X strings and energy

The X-string observable of length ℓ is the translated product

$$X_{\ell,N} = \frac{1}{N}\sum_{i=1}^{N}\left\langle\prod_{a=0}^{\ell-1} X_{i+a}\right\rangle, \quad \widehat{X}^{(k)}_{\ell,N} = \frac{1}{|\mathcal{B}_k|N}\sum_{t\in\mathcal{B}_k}\sum_i\prod_{a=0}^{\ell-1}\widehat{X}_{t,i+a}. \tag{S21}$$

The estimate is $\mathrm{median}_k\, \widehat{X}^{(k)}_{\ell,N}$. Supplementary Fig. 1 evaluates $\ell = 1, \ldots, 5$, while the decoding comparison uses $\ell = 1, 3, 5$. These are unconnected Pauli-string expectations, without an additional order-parameter rescaling.

For the TFI model, the total energy estimate is constructed by summing the Hamiltonian terms within each group:

$$\widehat{E}^{(k)}_{\mathrm{TFI}} = -\frac{1}{|\mathcal{B}_k|} \sum_{t \in \mathcal{B}_k} \sum_i \left[(1-g)\widehat{Z}_{t,i}\widehat{Z}_{t,i+1} + g\widehat{X}_{t,i} \right], \qquad \widehat{E}_{\mathrm{TFI}} = \mathrm{median}_k\, \widehat{E}^{(k)}_{\mathrm{TFI}}. \tag{S22}$$

The energy density is $\widehat{e} = \widehat{E}/N$. The same within-group Hamiltonian sum is used if J_1–J_2 or ANNNI energies are evaluated: the group-level spin-dot correlations or ZZ correlations and X mean are combined with the coefficients in Eqs. (S2) and (S3), multiplied by N, and then aggregated. Energy is not reconstructed by combining separately reported MoM observables.

S4 Rydberg reconstruction and MPS conditional calculations

Encoding, reference states and held-out records

Rydberg measurement records contain occupations $n_i \in \{0, 1\}$, where $n_i = 1$ denotes an excitation. In the MPS representation, physical index $b_i^{\mathrm{MPS}} = 0$ corresponds to an occupied site and $b_i^{\mathrm{MPS}} = 1$ to an empty site. Occupations are therefore obtained as $n_i = 1 - b_i^{\mathrm{MPS}}$ before evaluating physical observables. All measurement-basis tokens supplied to Diffushadow are fixed to Z. The eigenvalue convention in Section S3 applies to local-Pauli records; Rydberg statistics are evaluated directly from the occupations n_i.

Reference states are obtained by finite open-chain DMRG, and measurement records are sampled from their MPS Born distributions [4, 7]. Ground-state calculations use two-site updates with a mixer, a singular-value threshold of 10^{-10}, an energy convergence threshold of 10^{-9} and at most 40 sweeps. The finite-chain Hamiltonian and boundary conventions for both data generation and conditional-reference calculations are specified in Section S1.

Reconstruction is evaluated on independently sampled held-out records at $N = 32$. The initial evaluation grid contains $M_d = 1000$ records at each $d \in \{0.2, 1.8, 3.4, 5.0\}$, generated with sampling seed 4242. The extended grid contains $M_d = 500$ records at each $d \in \{4.0, 4.4, 4.8, 5.2, 5.6, 6.0\}$.

For each record s, the missing-site set $\mathcal{M}_s$ is sampled uniformly without replacement from all subsets containing $k = \max\{1, \mathrm{round}(fN)\}$ sites. Its complement $\mathcal{O}_s$ contains the observed sites. At $N = 32$, the nominal loss setting $f = 0.3$ gives exactly $k = 10$ missing sites per record. Conditional sampling, deterministic decoding, unconditioned fill and the Bayes calculation use the same intact records and per-record missing-site masks.

Conditional completion and controls

For each record s, only outcome tokens at sites in $\mathcal{M}_s$ are initialized to `[MASK]`. At each update, Diffushadow predicts the outcome distributions at all remaining missing sites, selects the site with the lowest predictive entropy, and fills its outcome token. Observed outcomes and previously completed tokens remain fixed. The updated record is then evaluated again, giving exactly k updates for a record with k missing sites.

For ensemble reconstruction, the selected outcome is sampled from the predicted distribution at temperature $\tau = 1$, with one completion per intact record. For missing-site accuracy, the selected outcome is assigned by stepwise argmax. The unconditioned-fill control generates a

complete configuration conditioned on d and the fixed measurement basis alone. Its occupations are inserted at $\mathcal{M}_s$, while the original observed occupations are retained. For the site-density accuracy baseline, the reference MPS marginal $\rho_j(d) = \langle n_j \rangle$ gives the prediction $\mathbf{1}[\rho_j(d) > 1/2]$ at each missing site. The unconditioned-fill control evaluates the contribution of record-specific conditioning to ensemble reconstruction, whereas the site-density baseline evaluates its contribution to missing-site prediction.

For an ensemble of M_d records, we define $\widehat{\rho}_i = M_d^{-1} \sum_s n_i^{(s)}$ and $\widehat{G}_{ij} = M_d^{-1} \sum_s n_i^{(s)} n_j^{(s)}$. The connected density structure factor is

$$\widehat{S}(q) = \frac{1}{N} \sum_{i,j=1}^{N} \cos[q(i-j)] \left(\widehat{G}_{ij} - \widehat{\rho}_i \widehat{\rho}_j \right). \tag{S23}$$

The sum uses open-chain site coordinates and includes the diagonal contribution $\widehat{\rho}_i(1-\widehat{\rho}_i)$. The covariance is evaluated with the $1/M_d$ ensemble normalization. Structure factors are evaluated on the grid $q_m = m\pi/60$, $m = 0, \ldots, 60$, which includes $q = 2\pi/3$. The same grid is used for the MPS reference and reconstructed ensembles.

Conditional marginals from the reference MPS

For the observed occupations in record s, define the projector

$$\Pi_s = \prod_{i \in \mathcal{O}_s} |n_i^{(s)}\rangle\langle n_i^{(s)}|. \tag{S24}$$

The reference conditional excitation probability at a missing site is

$$p_{s,j} = \frac{\langle \psi_d | \Pi_s n_j \Pi_s | \psi_d \rangle}{\langle \psi_d | \Pi_s | \psi_d \rangle}, \qquad j \in \mathcal{M}_s. \tag{S25}$$

The denominator is the probability of the observed pattern after summing over all missing occupations. The reference conditions only on the original observed outcomes; occupations generated during completion are not included in Π_s.

We evaluate these marginals by contracting the open-chain MPS in a double-layer representation. Let A_i^ν denote its local tensors, with physical index $\nu = 0$ corresponding to an occupied site and $\nu = 1$ to an empty site. For each record, define

$$w_{s,i}(\nu) = \begin{cases} \mathbf{1}[\nu = 1 - n_i^{(s)}], & i \in \mathcal{O}_s, \\ 1, & i \in \mathcal{M}_s. \end{cases} \tag{S26}$$

The physical index is therefore fixed at observed sites and summed over at missing sites. Starting from a scalar open-boundary environment, the left contraction obeys

$$L_{s,i+1}(c,d) = \sum_{a,b,\nu} L_{s,i}(a,b)\, w_{s,i}(\nu) (A_i^\nu)_{ac} (A_i^\nu)^*_{bd}. \tag{S27}$$

Right environments are contracted analogously from the opposite end. If $L_{s,j}$ and $R_{s,j}$ exclude site j, its two unnormalized weights are

$$Q_{s,j}(\nu) = \sum_{a,b,c,d} L_{s,j}(a,b) (A_j^\nu)_{ac} (A_j^\nu)^*_{bd} R_{s,j}(c,d), \qquad p_{s,j} = \frac{Q_{s,j}(0)}{Q_{s,j}(0) + Q_{s,j}(1)}. \tag{S28}$$

Reference MPSs are brought to canonical form before contraction. The reference MPSs have maximum bond dimensions ranging from 26 to 100, with a singular-value cutoff of 10^{-10} for

SVD truncation. The resulting conditional probabilities are those of the numerical reference MPS.

The expected Bayes-optimal missing-site accuracy is

$$A_{\mathrm{Bayes}}(d) = \frac{1}{M_d k} \sum_{s=1}^{M_d} \sum_{j \in \mathcal{M}_s} \max\{p_{s,j}, 1 - p_{s,j}\}. \tag{S29}$$

This quantity maximizes the expected number of correct individual site predictions under the observed information. It does not require identifying the most probable joint configuration of all missing sites. Model accuracy is measured against the withheld occupations in the intact records, whereas Eq. (S29) gives an expected accuracy. Finite-sample model accuracies may therefore lie slightly above the reference expectation.

As a numerical check, conditional marginals with no observed sites are compared with the density profile obtained directly from DMRG. The maximum absolute discrepancy in the reported reference calculations is of order 10^{-16}. Calculations were retained only when this discrepancy was below 5×10^{-3}. For numerical evaluation of Eq. (S28), the real parts of the contracted weights are normalized by their sum, without denominator regularization or probability clipping.

Blockade and multisite diagnostics

For each record s, let $\mathcal{M}_s$ denote the set of missing sites. A nearest-neighbour pair crosses the observed/reconstructed boundary when exactly one of its sites belongs to $\mathcal{M}_s$. We therefore define $a_{s,i} = |\mathbf{1}[i \in \mathcal{M}_s] - \mathbf{1}[i+1 \in \mathcal{M}_s]|$. For M_d records at detuning d, the boundary blockade-violation rate is

$$V_{\mathrm{boundary}} = \frac{\sum_{s=1}^{M_d} \sum_{i=1}^{N-1} a_{s,i} n_i^{(s)} n_{i+1}^{(s)}}{\sum_{s=1}^{M_d} \sum_{i=1}^{N-1} a_{s,i}}. \tag{S30}$$

The site sum follows the open-chain geometry. The denominator counts all scored boundary pairs across records, and the numerator counts those in which both sites are excited. The same missing-site masks define the scored pairs for intact records, conditional completion and unconditioned fill. Reported percentages are $100 V_{\mathrm{boundary}}$.

To assess multisite statistics, we evaluate period-three occupation products at orders $\ell = 2$ and $\ell = 8$. Their spatially averaged correlator is

$$G_\ell = \frac{1}{N - 3(\ell-1)} \sum_{i=1}^{N-3(\ell-1)} \left\langle \prod_{m=0}^{\ell-1} n_{i+3m} \right\rangle, \tag{S31}$$

where the brackets denote an average over all M_d records for the intact, conditionally completed and unconditioned-fill ensembles. Only origins for which every site in the product lies within the open chain are included.

For the complete-case estimator, define $J_{s,i,\ell} = \prod_{m=0}^{\ell-1} \mathbf{1}[i + 3m \notin \mathcal{M}_s]$. At each origin i, the occupation product is averaged over records with $J_{s,i,\ell} = 1$, and the resulting origin-wise means are then averaged spatially. Origins with no usable records are assigned undefined means and omitted from this correlator average. The mean number of usable records is

$$\overline{M}_{\mathrm{usable},\ell} = \frac{1}{N - 3(\ell-1)} \sum_{i=1}^{N-3(\ell-1)} \sum_{s=1}^{M_d} J_{s,i,\ell}. \tag{S32}$$

This count is averaged over all valid origins, including any with zero usable records. A record may contribute at one origin and be excluded at another. When k missing sites are selected uniformly without replacement from N sites, the probability that a specified ℓ-site set remains fully observed is $\binom{N-\ell}{k}/\binom{N}{k}$.

S5 Attention analysis and causal-mask control

Analysis inputs and attention-weighted distance

The analysis in main-text Fig. 3 uses the round-1 TFI model, trained at $N = 8, 10, 12, 14$, with six layers, eight heads, hidden dimension 256 and dynamic RoPE with scaling factor 4.0. The checkpoint is evaluated with all outcome tokens set to `[MASK]`, while the Hamiltonian parameter and measurement bases remain available.

For each N, we sample 512 basis configurations with random seed 1234, choosing each P_i independently and uniformly from $\{X, Y, Z\}$. The same configurations are reused across the parameter grid $\mathcal{G}_{\mathrm{attn}} = \{0, 0.1, 0.3, 0.5, 0.7, 0.9, 1.0\}$. Panels a–e use $N = 14$; panel f compares $N = 6, 8, 10, 12$ and 14.

Let $A_{qk}^{(\ell,h)}$ be the post-softmax attention weight from query token q to key token k in layer ℓ and head h. Masked outcome tokens are the queries. Both P_i and b_i have site label i; the parameter token has no site label. For $\mathcal{S} = \{P_i, b_i\}_{i=1}^{N}$ and $d_N(i,j) = \min\{|i-j|, N-|i-j|\}$, define

$$D^{(\ell,h)}(N,g) = \left\langle \frac{\sum_{k\in\mathcal{S}} A_{qk}^{(\ell,h)} d_N(s(q), s(k))}{\lfloor N/2 \rfloor \sum_{k\in\mathcal{S}} A_{qk}^{(\ell,h)}} \right\rangle_{q,\mathbf{P}}. \tag{S33}$$

The denominator conditions the spatial measure on attention assigned to basis and outcome keys and normalizes the maximum periodic distance to one. Same-site basis and outcome keys are included with zero distance; the parameter key is excluded. The query-wise ratios are averaged over query sites and basis configurations, retaining a separate value for each layer–head pair. Ratios are not formed after pooling attention rows. Main-text Fig. 3c reports the parameter sensitivity

$$\Delta D^{(\ell,h)} = \max_{g\in\mathcal{G}_{\mathrm{attn}}} D^{(\ell,h)}(14,g) - \min_{g\in\mathcal{G}_{\mathrm{attn}}} D^{(\ell,h)}(14,g), \tag{S34}$$

where the extrema are taken over the scanned parameter grid $\mathcal{G}_{\mathrm{attn}}$. Layer and head indices are zero-based; the six layers and eight heads are indexed by $\ell = 0, \ldots, 5$ and $h = 0, \ldots, 7$, respectively.

Site kernels and physical-correlation comparison

The signed displacement $\delta_N(i,j)$ is obtained from $(j-i) \bmod N$ and shifted into $(-N/2, N/2]$. For even N, the antipodal separation is represented once, at $+N/2$. The head-resolved kernels are

$$\begin{aligned} K_{bb}^{(\ell,h)}(\delta; N, g) &= \left\langle \sum_{j:\delta_N(i,j)=\delta} A_{b_i,b_j}^{(\ell,h)} \right\rangle_{i,\mathbf{P}}, \\ K_{bP}^{(\ell,h)}(\delta; N, g) &= \left\langle \sum_{j:\delta_N(i,j)=\delta} A_{b_i,P_j}^{(\ell,h)} \right\rangle_{i,\mathbf{P}}. \end{aligned} \tag{S35}$$

These averages retain the original attention weights; rows are not first renormalized within a key type. The heatmaps in main-text Fig. 3a,b additionally average the kernels over all layers and heads. They display the attention mass directed to each token type without separately renormalizing the two kernels. Kernels remain separate for each layer–head pair in the correlation-profile comparisons below.

For the comparison with physical correlations, the outcome-directed kernel is folded onto absolute periodic distance by

$$K_{bb}^{(\ell,h),\mathrm{abs}}(r) = \sum_{\delta:|\delta|=r} K_{bb}^{(\ell,h)}(\delta). \tag{S36}$$

Opposite displacements are summed. The self term and, for even N, the antipodal term are each counted once. The comparison uses the nonzero distances $\mathcal{R}_N$ available in both this kernel and the finite-size reference profile $C^{\mathrm{abs}}_{ZZ,N}(r,g) = |N^{-1}\sum_i \langle Z_i Z_{i+r}\rangle|$. The example in main-text Fig. 3d uses $N = 14$, $g = 0.7$ and $r = 1,\ldots,5$. The comparison ranges are $\mathcal{R}_6 = \{1,2,3\}$, $\mathcal{R}_8 = \{1,2,3,4\}$ and $\mathcal{R}_{10} = \mathcal{R}_{12} = \mathcal{R}_{14} = \{1,2,3,4,5\}$. Each size uses its corresponding finite-size exact ZZ correlation profile, matched to $\mathcal{G}_{\mathrm{attn}}$ by one-dimensional linear interpolation on the dense exact-reference g grid. Only points within the exact-grid range are accepted, with endpoint tolerance 10^{-6}.

Each profile is normalized by its own sum over $\mathcal{R}_N$, giving $\widetilde{K}^{(\ell,h)}_{bb}(r)$ and $\widetilde{C}_{ZZ,N}(r)$. The similarity is

$$S^{(\ell,h)}_{\cos}(N,g) = \frac{\sum_{r\in\mathcal{R}_N} \widetilde{K}^{(\ell,h)}_{bb}(r)\widetilde{C}_{ZZ,N}(r)}{\sqrt{\sum_{r\in\mathcal{R}_N} \widetilde{K}^{(\ell,h)}_{bb}(r)^2}\sqrt{\sum_{r\in\mathcal{R}_N} \widetilde{C}_{ZZ,N}(r)^2}}. \tag{S37}$$

Profiles with non-positive or non-finite total weight are assigned undefined normalized values. Cases for which the normalized profiles or cosine similarity are undefined are excluded from the summary statistics. The kernel is first averaged over query sites and basis configurations for each layer–head pair, and the cosine is computed separately for each pair. Main-text Fig. 3d displays L5H4, the head with the largest similarity among all 48 layer–head pairs at $N = 14$ and $g = 0.7$, with $S_{\cos} = 0.975$. Its selection is descriptive and is not an independently chosen test of alignment.

In main-text Fig. 3e, each point is the median similarity over the eight heads within one layer at $N = 14$, evaluated at $g = 0.3, 0.5, 0.7$ and 0.9. The shading spans the corresponding 25th and 75th percentiles. In main-text Fig. 3f, the median and interquartile range are taken over all 48 layer–head pairs at each N and g, for $N = 6, 8, 10, 12$ and 14. These summaries aggregate head-resolved similarities rather than computing the similarity of a kernel averaged across heads. The ranges describe variation among heads within the specified groups, not confidence intervals from repeated training or independent data samples. Parameter sensitivity and correlation-profile similarity measure distinct properties. Attention alignment alone does not establish that a particular head causes accurate observable generation or cross-scale transfer.

Causal-mask fixation control

Extended Data Fig. 1 compares bidirectional and causal information propagation at $g = 0.5$ and $N = 10$. The causal variant is obtained by replacing full attention with causal attention while retaining the trained weights, embedding layers and decoding strategy.

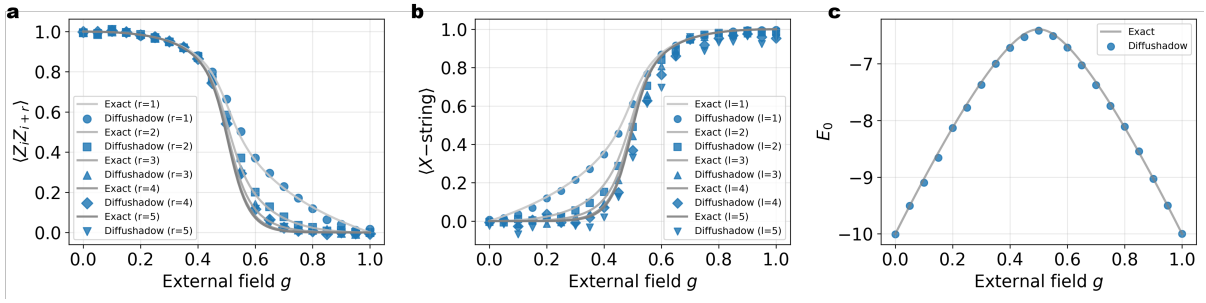
At every site, the measurement basis is sampled independently and uniformly from $\{X, Y, Z\}$, and all basis tokens are known. In the reference condition, all outcome tokens are masked. In the fixation condition, the outcome at a selected site j is fixed to $b_j = 1$, while all other outcomes remain masked. The measurement basis at site j remains randomly sampled. Each group contains 10,000 inputs and is evaluated once. Basis configurations are generated independently for each group.

The fixed site is varied over all ten sites. Target-site distributions under the reference and fixation conditions are compared for $i \neq j$, giving 90 off-diagonal comparisons for each attention variant. For binary distributions p_i and q_i, the KL divergence is

$$D_{\mathrm{KL}}(p_i \| q_i) = \sum_{b=0}^{1} p_i(b) \log \frac{p_i(b)}{q_i(b)}. \tag{S38}$$

The displayed matrix uses fixed site j as the row and target site i as the column, with diagonal entries excluded. Full attention allows the fixation to influence predictions at both earlier and

later sequence positions, whereas causal attention primarily propagates the change to later positions. This control illustrates the directional support of information propagation under the two attention masks.



Supplementary Fig. 1 | Additional observable estimation in the transverse-field Ising model. Ground-state observables of the 10-qubit transverse-field Ising model estimated from Diffushadow-generated classical-shadow measurements. (a) Two-point spin correlations $\langle Z_i Z_{i+r} \rangle$ for different correlation distances r. (b) X-string order parameters with different string lengths ℓ. (c) Ground-state energy E_0. Grey solid curves denote exact diagonalization results, and blue markers denote estimates obtained from Diffushadow-generated classical-shadow measurements. In panels a and b, marker shapes distinguish the correlation distance r and the string length ℓ, respectively. These additional observables complement the longitudinal spin-correlation results emphasized in the main text and show that Diffushadow-generated shadows support estimation of both local correlations, extended string observables and the ground-state energy.

S6 Additional finite-size observable validation

TFI correlations, strings and energy

Supplementary Fig. 1 tests observables beyond the representative TFI correlations emphasized in the main text. At $N = 10$, generated Pauli records are used to estimate ZZ correlations at $r = 1, \ldots, 5$, X strings of length $\ell = 1, \ldots, 5$, and the total energy E_0. The operators, spatial averages and MoM order are defined in Section S3. The evaluation uses 15 uniformly spaced values of $g \in [0, 1]$ and $M = 10{,}000$ records at each value.

The generated-data estimates follow the ED reference trends: longitudinal correlations decrease as the transverse field grows, while the displayed X-string expectations increase. The energy also follows the reference curve. This comparison supports agreement for the selected local, extended-string and Hamiltonian observables at the tested size. It does not establish equality of the complete measurement distributions. The decoding comparisons for the string observables are discussed in Section S7 rather than repeated here.

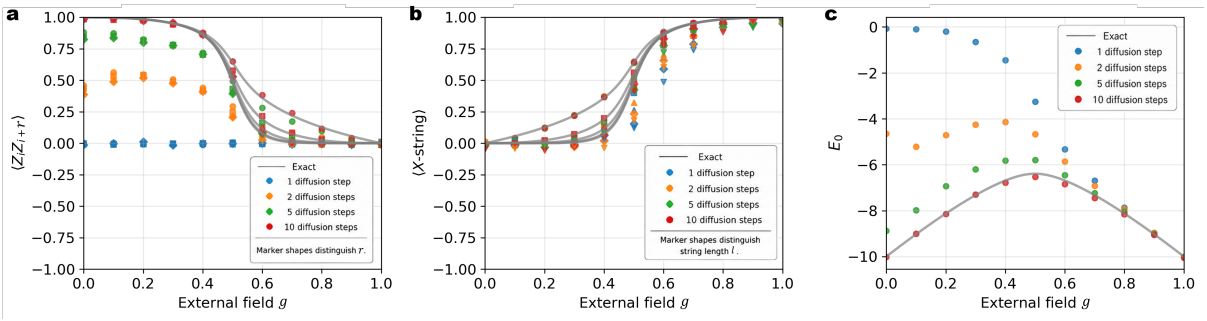
S7 Dependence on decoding strategy and sampling budget

Decoding strategy and number of updates

We compare decoding protocols on the $N = 10$ TFI benchmark using the observable definitions in Section S3. Entropy-based decoding selects the remaining site or sites with the lowest predictive entropy; random-position decoding chooses among the remaining sites uniformly. One-shot decoding fills all outcomes from a single model evaluation of the fully masked sequence. The position-selection rule and the outcome-value rule are distinct settings.

Supplementary Fig. 2 compares $T = 1, 2, 5, 10$ updates. At $N = 10$, these settings reveal $10, 5, 2, 1$ sites per update, respectively, with predictions recomputed after each group of sites has been filled. The curves show the strongest one-shot deviations in the low-field correlation and energy estimates. More updates improve agreement for these observables by making newly generated outcomes available to later predictions. The improvement from additional denoising steps is especially visible in the low-field regime $g < 0.5$, where the transverse-field Ising ground state exhibits stronger longitudinal order and longer-range spin correlations. In this regime, accurately generating globally consistent measurement outcomes requires the model to capture correlations across multiple sites, and gradual denoising provides a useful mechanism for progressively establishing these correlations. By contrast, in the high-field paramagnetic regime $g > 0.5$, longitudinal correlations become weaker and more short-ranged, making the generated observables less sensitive to the number of refinement steps.

Supplementary Table 2 compares ShadowGPT, random-position decoding, one-shot decoding and entropy-based iterative decoding at $g = 0.1, 0.5, 0.9$. Diffu-rdm denotes iterative decoding with randomly selected positions, Diffu-s1 denotes one-step decoding, and Diffu denotes the full entropy-based iterative decoding strategy. The one-step variant shows substantially larger errors, especially for the ground-state energy, confirming the importance of iterative denoising. Random position selection can perform competitively for some individual observables, but is less consistent than the full entropy-based strategy. Overall, full Diffushadow achieves low errors across representative observables and provides a stable alternative to sequential autoregressive generation.



Supplementary Fig. 2 | Effect of denoising steps on observable estimation in the transverse-field Ising model. Physical observables of the 10-qubit transverse-field Ising model estimated from Diffushadow-generated classical-shadow measurements using different numbers of denoising steps. (a) Two-point spin correlations $\langle Z_i Z_{i+r}\rangle$. (b) X-string order parameters with different string lengths ℓ. (c) Ground-state energy per site E_0/N. Grey solid curves denote exact diagonalization results, and colored markers denote Diffushadow estimates generated with 1, 2, 5 and 10 denoising steps. In panel a, marker shapes distinguish the correlation distance r; in panel b, marker shapes distinguish the string length ℓ. Increasing the number of denoising steps improves the agreement between generated-measurement estimates and exact results, with one-step decoding showing the largest deviation and 10-step denoising giving the most accurate estimates.

Number of generated records

Supplementary Fig. 3 compares $M = 8000$, 10,000, 100,000 and 300,000 generated records per parameter point at $N = 10$. Each ensemble is processed with $K = 10$ groups; the resulting group sizes are 800, 1000, 10,000 and 30,000 records. The same ZZ correlations, X strings and total-energy observables are used. The displayed observable trends are already recovered at the smaller budgets and remain similar over the tested range. Larger ensembles produce smoother curves in these plots. However, even the smaller sampling budgets already recover the main observable trends, including the decay of longitudinal spin correlations, the growth of X-string

Supplementary Table 2 | MSE of selected observable estimates for the 10-qubit transverse-field Ising model. The table reports the ground-state energy E_0 and representative X-string observables; ZZ correlations are omitted to avoid redundancy with the main text. ShGPT denotes ShadowGPT, Diffu-rdm denotes Diffushadow with randomly selected decoding positions, Diffu-s1 denotes one-step Diffushadow decoding, and Diffu denotes the full Diffushadow with entropy-based iterative decoding.

	Model	E_0	$\left(\prod_{i=1}^{1} X_i\right)$	$\left(\prod_{i=1}^{3} X_i\right)$	$\left(\prod_{i=1}^{5} X_i\right)$
$g = 0.1$	ShadowGPT	3.63×10^{-4}	7.56×10^{-4}	6.80×10^{-4}	2.77×10^{-3}
	Diffu-rdm	2.34×10^{-3}	2.90×10^{-4}	1.88×10^{-6}	3.69×10^{-3}
	Diffu-s1	7.99×10^{1}	1.92×10^{-4}	1.16×10^{-4}	4.10×10^{-4}
	Diffushadow	$\mathbf{2.77\times10^{-4}}$	$\mathbf{2.20\times10^{-4}}$	$\mathbf{1.85\times10^{-7}}$	$\mathbf{1.33\times10^{-3}}$
$g = 0.5$	ShadowGPT	1.35×10^{-2}	6.31×10^{-5}	2.15×10^{-4}	2.77×10^{-4}
	Diffu-rdm	7.59×10^{-3}	2.17×10^{-4}	6.00×10^{-4}	1.12×10^{-2}
	Diffu-s1	1.00×10^{1}	1.05×10^{-4}	5.82×10^{-2}	1.71×10^{-1}
	Diffushadow	$\mathbf{7.13\times10^{-3}}$	$\mathbf{4.01\times10^{-5}}$	$\mathbf{4.90\times10^{-5}}$	$\mathbf{2.29\times10^{-4}}$
$g = 0.9$	ShadowGPT	6.79×10^{-4}	9.88×10^{-6}	4.76×10^{-5}	7.57×10^{-5}
	Diffu-rdm	5.05×10^{-4}	6.57×10^{-8}	6.20×10^{-4}	4.28×10^{-3}
	Diffu-s1	1.47×10^{-3}	4.44×10^{-6}	4.33×10^{-4}	1.21×10^{-3}
	Diffushadow	$\mathbf{4.98\times10^{-4}}$	$\mathbf{3.32\times10^{-6}}$	$\mathbf{2.22\times10^{-8}}$	$\mathbf{3.88\times10^{-4}}$

observables and the ground-state energy curve. These results suggest that the agreement with exact results is not solely a consequence of using an extremely large number of generated samples, but also reflects the quality of the measurement distribution learned by Diffushadow.

S8 Additional Rydberg reconstruction diagnostics

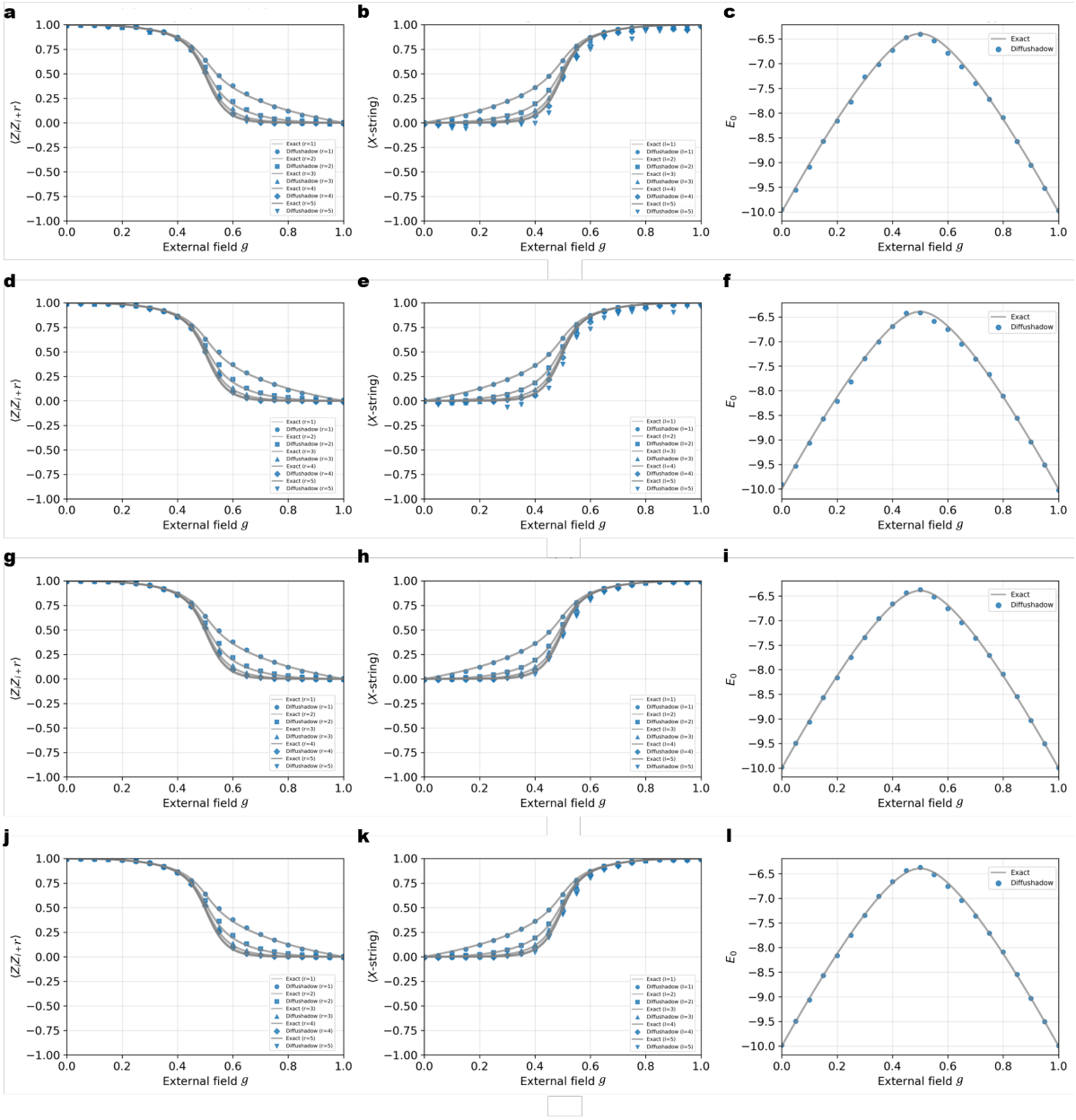
These diagnostics supplement the occupation structure factor and missing-site accuracy in Fig. 5. They use $N = 32$, $R_b/a = 2.35$, $f = 0.3$ and $k = 10$ uniformly chosen missing sites per record. The intact, conditional-sampling and unconditioned-fill ensembles share the same $M_d = 1000$ records and masks at each of the four original detunings. The definitions and comparison protocol are given in Section S4. Conditional ensemble results use probability sampling at $\tau = 1$, rather than the argmax completion used for site accuracy.

Consistency across observed and reconstructed neighbours

We evaluate nearest-neighbour double excitations across the boundary between observed and reconstructed sites. For each detuning, 10 sites are randomly masked in each of 1000 held-out records at $N = 32$. A neighbouring pair is included when exactly one of its sites is masked, giving approximately 1.38×10^4 scored pairs per detuning. The same pair sets are used for intact records, conditional completion and unconditioned fill (control). Counts and rates, computed using Eq. (S30), are listed below.

d	Intact count	Conditional count	Conditional rate	Control count	Control rate
0.2	0	1	0.007%	225	1.63%
1.8	0	0	0.000%	312	2.26%
3.4	0	1	0.007%	15	0.11%
5.0	0	2	0.014%	508	3.68%

Across the four detunings, conditional completion produces four doubly excited boundary pairs, compared with 1060 for unconditioned fill. The conditional rates remain below 0.02% at every detuning. These results support the role of conditioning on the observed occupations in preserving blockade consistency across the reconstruction boundary.



Supplementary Fig. 3 | Dependence of observable estimation on the number of generated classical-shadow snapshots in the transverse-field Ising model. Ground-state properties of the 10-qubit transverse-field Ising model estimated from Diffushadow-generated classical-shadow measurements using different numbers of generated snapshots M. Panels a–c, d–f, g–i and j–l correspond to M = 8k, 10k, 100k and 300k, respectively. In each row, the left panel shows the two-point spin correlations $\langle Z_i Z_{i+r} \rangle$, the middle panel shows the X-string order parameters $\langle X\text{-string} \rangle$ with different string lengths, and the right panel shows the ground-state energy per site E_0/N. Grey solid curves denote exact diagonalization results, and blue markers denote estimates obtained from Diffushadow-generated measurements. In the correlation and string-order panels, different marker shapes distinguish different correlation distances r and string lengths l, respectively. As the number of generated snapshots increases, the estimated observables remain consistently close to the exact results, indicating that Diffushadow provides stable physical-property estimation across different sampling budgets.

Multisite observables and complete-case sample counts

At $d = 5.0$, we evaluate the period-three products G_ℓ at $\ell = 2$ and $\ell = 8$ to assess whether conditional completion preserves multisite statistics. The estimates from intact records, conditional completion and unconditioned fill (control), together with the mean numbers of records available to the complete-case estimator, are listed below.

ℓ	Intact G_ℓ	Conditional G_ℓ	Control G_ℓ	Mean usable records
2	0.297	0.295	0.271	467
8	0.064	0.062	0.026	28

The complete-case counts are averaged over spatial origins, as defined in Eq. (S32), out of 1000 held-out records. A record contributes at a given origin only when all sites involved in the product are observed. Conditional completion uses all 1000 records at every valid origin.

For both orders, the conditional estimates closely reproduce the intact values, whereas unconditioned fill gives lower estimates. The reduction in complete-case counts from 467 at $\ell = 2$ to 28 at $\ell = 8$ illustrates the loss of usable records for higher-order observables. Conditional completion retains the full ensemble for estimating both products.

Excitation-number fluctuations

For each record s, we define the total excitation number as $Q_s = \sum_i n_i^{(s)}$. At $d = 5.0$, the mean excitation numbers are 10.72 for intact records and 10.68 for conditional completion. The population standard deviations across records are 0.521, 0.538 and 1.168 for intact records, conditional completion and unconditioned fill, respectively. Conditional completion closely reproduces both the mean and the fluctuations of the intact ensemble, whereas unconditioned fill produces substantially larger excitation-number fluctuations.

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